

NAVYASRI PULIPATI

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING STUDENT

+91-8904985309 | pulipatinavyasri@gmail.com

PROFILE

AIML undergraduate with strong foundations in software engineering and machine learning. Experienced in building scalable backend systems, cloud-native applications, and AI-powered solutions using Python, FastAPI, Docker, and Kubernetes.

EDUCATION

R.V. College of Engineering, Bangalore : Bachelor of Engineering in Artificial Intelligence and Machine Learning, 4th Year (current)
| **CGPA: 9.32/10** | Bangalore, Karnataka, India

WORK EXPERIENCE

HPE CPP3 Project Intern : HPE (6 months)

Contributed to a VS Code extension for HPE's PL/MX language, owning the Tree-sitter AST semantic analyzer, deterministic SQL test generation, and LSP features including diagnostics, hover, and completions.

AI Research Intern : Xtelify Limited (Airtel Digital) (6 months)

Engineered a 4-agent AI framework (predictive, optimization, anomaly, orchestrator) for autonomous telecom capacity management, automating real-time, predictive resource allocation.

Edge AI Intern : Aeronautical Development Agency (ADA) (4 weeks)

Built a real-time testing framework with custom-built GStreamer plugins for Automatic Target Recognition (ATR) systems at ADA using NVIDIA DeepStream SDK.

AI Research Intern : Anthrasync Solutions Private Limited (3 months)

Developed 3 core AI agents (Translation, Summarization, and Resource Finder) for the company's agentic AI platform, automating multilingual communication and intelligent web retrieval.

PROJECTS

Federated Learning MLOps Platform

Orchestrated a concurrent Federated Learning MLOps pipeline using Prefect and 16-worker thread pools, integrating a FastAPI and MLflow control plane with 5-second health checks and 30-minute timeouts for fault-tolerant experiment tracking.

Offline AI OS

Engineered a privacy-first offline AI platform using ONNX Runtime, Ollama, and ChromaDB, integrating local RAG, tool orchestration, and document processing while achieving 18.85 ms average CPU embedding latency on AMD Ryzen.

AI Token Optimizer

Developed a cloud-native AI Token Cost Predictor and Optimizer using Spark, FastAPI, Azure, and the ELK stack, supporting 15 LLMs across 3 providers. Achieved 20–26% token reduction and R^2 up to 0.994, with real-time token prediction and cost estimation.

RESEARCH & PUBLICATIONS

Pulipati, N. M. (First Author), "Music-Based Cryptography: Text Encryption Using Audio Features," *IEEE IC3IT 2025* : proposed an audio-feature-driven encryption framework using entropy optimization. DOI: 10.1109/IC3IT66137.2025.11341611

TECHNICAL SKILLS

Languages: Python, R, C, SQL | **Core CS:** DSA, OOP, Operating Systems, DBMS, Computer Networks

Backend: FastAPI, REST APIs, Redis | **DevOps/Cloud:** Docker, Kubernetes, Git, Linux, Azure

AI/ML: Scikit-learn, LangGraph, MLflow, Prefect, RAG, XGBoost

Soft Skills: Teamwork, Time Management, Leadership, Effective Communication, Critical Thinking

COURSES & CERTIFICATIONS

Data Science for Engineers : NPTEL, Elite+Gold (90%) | **Machine Learning** : Stanford/DeepLearning.AI | **Generative AI & Vertex AI** : Google Cloud